

SmartClass Vision — Complete Technical Report

System Implementation, Model Justification & Architecture Summary

1. Executive Overview & What Was Implemented

SmartClass Vision is an automated classroom attendance and security intelligence system designed to eliminate attendance fraud, proxy check-ins, and manual roll-call delays. The system captures synchronized burst frames of the classroom, detects and aligns all faces simultaneously, validates biometric quality and liveness, matches identities using high-dimensional angular embeddings, and cryptographically signs audit logs.

Implemented Feature	Technical Functionality & Engineering Implementation
OpenCV YuNet Detection & Landmark Alignment	Replaced generic detectors with YuNet C++ ONNX model. Detects faces at micro-scales and extracts 5 landmarks (eyes, nose, mouth) to mathematically rotate faces via Affine Transformations before recognition.
InsightFace ArcFace 512-D Recognition	Upgraded legacy FaceNet (128-D) to ArcFace (512-D). Employs Additive Angular Margin Loss to maximize inter-person separation. Feature vectors are L2-normalized and matched via vectorized cosine distance.
Multi-Shot Temporal Voting Consensus	Captures a 10-shot burst across the classroom. Enforces a minimum 5-shot quorum and average cosine score ≥ 0.65 to prevent false triggers from passers-by outside doors.
Strict Section & Class Isolation	Validates candidates against target Degree, Year, Branch, and Section. Students enrolled in another class are explicitly flagged REJECTED - WRONG CLASS , logged as incidents, and denied attendance.
Hardware-Independent Anti-Spoofing	Multi-stage quality firewall checks Laplacian sharpness (≥ 40.0), pose yaw ratio (≤ 1.4), and high-frequency Moiré patterns to reject smartphone screen and printed photo presentation attacks.
Cryptographic Signatures (HMAC-SHA256)	Every generated attendance CSV is hashed and cryptographically signed with the teacher's credentials using HMAC-SHA256, permanently preventing manual offline alterations in Excel.
5-Pose Guided Student Registration HUD	Interactive registration camera interface requiring 5 distinct angles (Center, Left, Right, Up, Down) with live quality meters (sharpness, light) and instant Spacebar capture lock.

2. Which Models Are Being Used & Why

The project strictly runs two modern, state-of-the-art computer vision models. All obsolete legacy models (such as YOLOv8-Face, FaceNet 128-D, and dlib) were completely removed:

MODEL 1: OpenCV YuNet (Face Detection & Alignment)	MODEL 2: InsightFace ArcFace (Face Recognition)
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• Model File: face_detection_yunet_2023mar.onnx • Execution Engine: Native OpenCV C++ DNN module. • Input/Output: Full frame → BBoxes + 5-point facial landmarks. • Why YuNet? 1. Extremely Fast: Runs in ~15ms on standard CPU hardware without GPU requirements. 2. Micro-Face Detection: Accurately detects distant students in large lecture halls. 3. Built-in Landmark Regression: Provides coordinates for both eyes, nose, and mouth corners, enabling precise Affine transformation alignment so faces are upright and standardized before recognition.	• Model File: w600k_r50.onnx (ResNet-50 backbone). • Execution Engine: ONNX Runtime / Vectorized NumPy. • Embedding Size: 512-Dimensional L2-Normalized Vector. • Why ArcFace over FaceNet / dlib? 1. Additive Angular Margin: ArcFace optimizes geodesic distance on a hypersphere. Unlike FaceNet's Euclidean triplet loss, ArcFace produces compact intra-class clusters and huge inter-class margins. 2. Extreme Accuracy: Achieves >99.8% on LFW benchmark. 3. Zero Ambiguity: Enforces dual thresholds (Cosine ≥ 0.65 and Top-1 vs Top-2 Margin ≥ 0.12).
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3. Core Numerical Parameters & Thresholds (To Remember)

Parameter	Threshold	Operational Meaning & Engineering Defense
ArcFace Cosine Similarity	≥ 0.65	Confidence requirement for biometric identity confirmation.
Top-1 vs Top-2 Margin	≥ 0.12	Ambiguity gate. Discards matches if the top two candidates are too close.
Burst Capture Shots	10 Frames	Synchronized high-resolution bursts across the classroom.
Temporal Quorum Votes	≥ 5 of 10	Candidate must be detected and verified across at least 5 frames.
Min Sharpness (Laplacian)	40.0	Rejects motion-blurred or poorly focused face crops.
Max Pose Yaw Ratio	≤ 1.40	Rejects extreme profile/side-facing head angles.
Registration Poses	5 Angles	Center, Left, Right, Up, Down for complete facial feature enrollment.

Summary for Viva / Presentation: SmartClass Vision pairs **OpenCV YuNet** (for 15ms 5-landmark affine face alignment) with **InsightFace ArcFace** (for 512-D angular margin biometrics). Attendance is proxy-proof via 10-shot temporal consensus, strict Section isolation, liveness anti-spoofing, and tamper-evident HMAC-SHA256 signatures.